CYBERSECURITY INTERNSHIP ASSESSMENT

Network Scanning & Topology Technical Report

Module 05: Active Host Discovery & LAN Enumeration with Zenmap

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Authorization Ref	NW-LOA-B084C-023	**Target Subnet**	10.57.180.0/24
Assessment Type	Active Ping Sweep & Topology	**OS / Scanner**	Windows / Zenmap (Nmap 7.993)

1. EXECUTIVE SUMMARY & SCOPE

This technical report documents the active network scanning and host discovery assessment performed on local LAN subnet **10.57.180.0/24** under authorization ref **NW-LOA-B084C-023**. Utilizing **Zenmap (Nmap 7.993)** on a Windows host, seven structured lab tasks were completed to identify local network parameters, discover live endpoints, enumerate hardware MAC addresses, and export a graphical network topology map.

2. TASK 1 & 2: ZENMAP INSTALLATION & LOCAL SUBNET DISCOVERY

Technical Assessment: Installed the official Nmap `7.993` Windows binary bundle including the Zenmap GUI frontend and Npcap packet capture driver. Querying the host network configuration via `ipconfig` identified the active **Wireless LAN adapter Wi-Fi** interface at `10.57.180.168` with a `/24` subnet mask, establishing the target scan range **10.57.180.0/24** and default gateway **10.57.180.174**.

```
C:\> ipconfig
```

```
Wireless LAN adapter Wi-Fi: IPv4 Address. . . . . . . . . . . : 10.57.180.168
Subnet Mask . . . . . . . . . . . : 255.255.255.0 Default Gateway . . . . . . . . .
: fe80::480f:fff:fecd:efd8%16 10.57.180.174 Calculated Target LAN Subnet . . :
10.57.180.0/24
```

3. TASK 3: ACTIVE HOST DISCOVERY PING SCAN (ZENMAP)

Technical Assessment: Configured Zenmap with target `10.57.180.0/24` and selected the **Ping scan** profile (`nmap -sn`) to perform host discovery without intrusive port probing. Nmap swept all 256 IPv4 addresses in **3.54 seconds** and resolved active ARP/ICMP responses.

```
$ nmap -sn 10.57.180.0/24
```

```
Starting Nmap 7.993 ( https://nmap.org ) at 2026-11-13 16:50 +0530 Nmap scan report
for 10.57.180.21 Host is up (0.026s latency). MAC Address: 6C:E8:C6:E5:F4:22 (Earda
Technologies) Nmap scan report for 10.57.180.174 Host is up (0.022s latency). MAC
Address: 4A:0F:0F:CD:EF:D8 (Unknown) Nmap scan report for 10.57.180.168 Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.54 seconds
```

4. TASK 4, 5 & 6: LIVE HOSTS, IP & MAC ADDRESS ENUMERATION

Technical Assessment: A total of **3 hosts are live** in the `10.57.180.0/24` subnet (including the local scanning PC). Hardware MAC addresses and OUI vendor mappings were extracted for remote nodes, while the local scanning adapter MAC is retrieved directly via `ipconfig /all`.

NO.	LIVE IP ADDRESS	MAC ADDRESS	LATENCY	VENDOR / ROLE
01	`10.57.180.21`	`6C:E8:C6:E5:F4:22`	0.026s	Earda Technologies (Client)
02	`10.57.180.174`	`4A:0F:0F:CD:EF:D8`	0.022s	Default Gateway / Hotspot
03	`10.57.180.168`	Local Wi-Fi Adapter (`ipconfig /all`)	Local	Local Scanning Windows PC

5. TASK 7: NETWORK TOPOLOGY VISUALIZATION & PDF EXPORT

Technical Assessment: Opened the **Topology** tab in Zenmap, reviewed the host and connection legend, and exported the network map in PDF format via **Save Graphic**. The radial graph maps `localhost` (`10.57.180.168`) alongside active peer nodes `10.57.180.21` and gateway `10.57.180.174`.

```
Zenmap GUI > Topology Tab → Save Graphic (Format: PDF)
```

```
Exported Topology Nodes: localhost | 10.57.180.168 | 10.57.180.174 | 10.57.180.21
Connection State: Direct LAN Subnet Links Verified (No Intermediate Traceroute Hops)
```

6. NETWORK RECONNAISSANCE SUMMARY & EXPOSURE MATRIX

NETWORK COMPONENT	DISCOVERED DATA / METRIC	SECURITY & ATTACK SURFACE IMPLICATION
LAN Subnet Scope	`10.57.180.0/24` (256 IPs)	Flat local broadcast domain allows rapid ARP-based host enumeration.
Gateway Node	`10.57.180.174` (`4A:0F:0F:CD:EF:D8`)	Primary routing gateway using locally administered/randomized MAC address.
Active Peer Host	`10.57.180.21` (Earda Tech)	Live endpoint reachable on LAN; candidate for service/port enumeration in Phase 3.
Discovery Footprint	3 Hosts Up in 3.54s (`-sn`)	Ping sweep completed rapidly without triggering host-level firewall blocks.

7. ASSESSMENT CONCLUSION

The active network scanning phase successfully mapped the **10.57.180.0/24** subnet, identifying 3 live endpoints, resolving MAC hardware identifiers, and generating a visual topology diagram for subsequent vulnerability analysis. All scanning activities were confined strictly to the authorized local network environment.